

Reader: The limit of the function $\sqrt{1-x^2}$ at $x=0$ exists and is equal to zero. The limit of the function $\sqrt{x-1}$ at $x=1$ also exists and is also equal to zero. According to the theorem on the limit of the sum, the limit of $f(x) = \sqrt{1-x^2} + \sqrt{x-1}$ must exist and be equal to the sum of the two preceding limits, i.e. to zero.

Author: Nevertheless, $f(x) = \sqrt{1-x^2} + \sqrt{x-1}$ has no limit at $x=1$ for a simple reason that the expression $(\sqrt{1-x^2} + \sqrt{x-1})$ has meaning only at a single point (point $x=1$). Applying the theorem on the limit of the sum, you have not taken into account the domains of the functions $\sqrt{1-x^2}$ and $\sqrt{x-1}$. The former has the natural domain over $[-1, 1]$, while the latter over $[1, \infty[$.

Reader: Apparently your note also covers the cases when the theorems on the limit of the product and the limit of the ratio of functions are used.

Author: It goes without saying. Working with functions, you must always consider their domains. The natural domains of functions may intersect (or even coincide), but sometimes they may not. This aspect must never be overlooked. Why do you think we never have such complications when working with sequences?

Reader: Obviously because all numerical sequences have one and the same domain, i.e. a set of natural numbers.

Author: Correct. Now we come to Note 2. Do you think the limit $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ exists?

Reader: In any case the theorem on the limit of the ratio is not valid here because $\lim_{x \rightarrow 0} x = 0$.

Author: In fact, if $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$, the limit of their ratio i.e. the limit of the function $(\frac{f(x)}{g(x)})$,